Beyond rectangles: a candidate class $\mathcal{C}$

a positive case for the two monotonicities, and the one lemma that closes it

Setup. As before, $X \sim N_2(\mu, I_2)$,

$$\Lambda = \{\mu : \mu_2 \le \mu_1\}, \qquad \Lambda_0 = \{\mu \in \Lambda : \mu_1 \le 0\}, \qquad \Lambda_1 = \{\mu \in \Lambda : \mu_1 > 0\}.$$

A nonrandomized test is $\phi = \mathbf{1}_{A^c}$ with acceptance region A. Write $\sigma(x) := (x_2, x_1)$ for the swap, $a^{\downarrow} := (\max(a_1, a_2), \min(a_1, a_2))$ for the descending sort, and $h_A(a) := \sup_{x \in A} a'x$ for the support function. Power is $\beta_\phi(\mu) = 1 - M_\phi(\mu)$ with miss probability $M_\phi(\mu) = P_\mu(X \in A)$.

1. The class

Definition 1 (the two monotonicities)**.** A test $\phi = \mathbf{1}_{A^c}$ belongs to $\mathcal{C}$ if its acceptance region satisfies:

(M1) *Coordinatewise monotonicity*: A is a lower set, i.e. $x \le y$ and $y \in A$ imply $x \in A$. Equivalently ϕ is nondecreasing in each coordinate.

(M2) *Order-respecting priority of coordinate* 1: if $x_1 \ge x_2$ and $x \in A$, then $\sigma x \in A$. Equivalently $\phi(x_1, x_2) \ge \phi(x_2, x_1)$ whenever $x_1 \ge x_2$.

We write $\mathcal{C}_{\le\alpha}$ for the members of level $\le \alpha$, and $\mathcal{C}_\alpha$ for those of exact size α.

The maintained test will, in addition, have a *closed convex* acceptance region. The ordered rectangles of the previous section are exactly the rectangular members of $\mathcal{C}$.

Conjecture 1. *If $\phi \in \mathcal{C}_\alpha$ has a closed convex acceptance region, then ϕ is admissible relative to $\mathcal{C}_{\le\alpha}$.*

I believe this. Convexity is needed only for the maintained test; the competitor is asked only for (M1)–(M2). The rest of this note is the affirmative case: five reasons, organized so that the last one names the single lemma whose proof would finish the theorem.

2. Why these two conditions, and no others

The restricted problem manufactures dominators in exactly two ways, and (M1)–(M2) are calibrated one-to-one against them.

- **Sign-cancellation** (a test that rejects on a thin oblique band, e.g. $\phi_\delta = \mathbf{1}\{x_2 - x_1 > c\}$, aiming mass at the deleted region $\mu_2 > \mu_1$). This is killed by (M1): ϕ_δ is not monotone, since raising x_1 can switch rejection off.
- **Priority inversion** (a monotone test that spends its budget on the low-priority coordinate, e.g. $\mathbf{1}\{x_2 > q_\alpha\}$). This is killed by (M2): it rejects when the large value sits in coordinate 2, and is itself dominated by the coordinate-1 test.

The reason I expect the list to be *complete* is that (M1)–(M2) are precisely the fixed points of the two power-improving rearrangements available on Λ_1. On Λ_1 coordinate 1 carries the binding mean ($\mu_1 \geq \mu_2$, $\mu_1 > 0$). Monotonizing a test (taking the lower-set hull of its acceptance region) and then sorting it toward coordinate 1 (the swap fold of (M2)) each *weakly raise* power on Λ_1, because both move rejection mass toward the informative coordinate. A test is in $\mathcal{C}$ iff it is a fixed point of both. So $\mathcal{C}$ is not a convenience class carved to fit the answer: it is the set of tests that cannot be improved by the only two cost-free rearrangements the geometry of Λ_1 allows.

3. Support-sorting: every infeasible face has a feasible mirror

Call a direction a *feasible* if a ray $\theta^* + \lambda a$ ($\lambda \to \infty$) can be kept inside Λ_1; since $\mathrm{rec}(\Lambda_1) = \{d : d_2 \leq d_1\}$, feasibility is $a_2 \leq a_1$. The Lehmann–Romano ray argument runs along feasible directions and stalls on infeasible ones ($a_2 > a_1$) — this is the only reason convexity failed to give admissibility in the restricted problem. Condition (M2) repairs exactly this.

Lemma 1 (support-sorting)**.** *If A is order-respecting, then $h_A(a^{\downarrow}) \leq h_A(a)$ for every $a \in \mathbf{R}^2$.*

Proof. If $a_1 \geq a_2$ then $a^{\downarrow} = a$. Suppose $a_2 > a_1$, so $a^{\downarrow} = \sigma a$. Fix $x \in A$. If $x_1 \geq x_2$, then $\sigma x \in A$ by (M2), and $(\sigma a)'x = a_2 x_1 + a_1 x_2 = a'(\sigma x) \leq h_A(a)$. If $x_1 < x_2$, then $(\sigma a)'x - a'x = (a_2 - a_1)(x_1 - x_2) < 0$, so $(\sigma a)'x < a'x \leq h_A(a)$. Taking the supremum over $x \in A$ gives $h_A(a^{\downarrow}) \leq h_A(a)$. □

The consequence: if the maintained convex body A is bounded above in any infeasible direction a, it is bounded above in the feasible mirror $a^{\downarrow}$ as well, with no larger reach. Since (M1) forces the supporting directions of A to be nonnegative ($a \geq 0$), and feasibility adds $a_2 \leq a_1$, the *feasible supporting cone* of any test in $\mathcal{C}$ is

$$\mathcal{F} := \{a : 0 \leq a_2 \leq a_1\} = \mathrm{cone}\{(1,0), (1,1)\}.$$

This is the clean structural picture: the directions in which we can probe the alternative are exactly the wedge between the coordinate-1 axis $(1,0)$ and the diagonal $(1,1)$, and Lemma 1 says nothing important about A lives outside this wedge that is not already pinned inside it.

4. The two probes are the two edges of $\mathcal{F}$

Recall the rectangle proof used two limits: $\mu_2 \to -\infty$ (interior) and $\mu = (t,t) \to \infty$ (diagonal). These are precisely the rays along the two extreme directions of $\mathcal{F}$:

$$(1,0) \rightsquigarrow \text{coordinate-1 cutoff} \quad c_1(\phi) := h_\phi\big((1,0)\big), \qquad (1,1) \rightsquigarrow \text{diagonal reach} \quad s(\phi) := h_\phi\big((1,1)\big).$$

Under (M1), the interior limit collapses *any* member of $\mathcal{C}$ to the univariate threshold $\mathbf{1}\{x_1 > c_1(\phi)\}$, so domination forces $c_1(\psi) \leq c_1(\phi)$ exactly as in Step 1. The diagonal limit is governed by $s(\cdot)$: along $\mu = (t,t)$,

$$\log M_\phi(t,t) = s(\phi)\, t - t^2 + O(\log t),$$

so domination forces $s(\psi) \le s(\phi)$.

More is true, and this is the general engine. Probing along an *arbitrary* $a \in \mathcal{F}$ (push μ to infinity along that ray) gives, for any dominating ψ,

$$h_\psi(a) \le h_\phi(a) \qquad \text{for every } a \in \mathcal{F}, \tag{1}$$

because the acceptance probability along the ray decays at the rate set by the reach in direction a, and the slower-decaying (larger-reach) test would have strictly larger acceptance far out. So domination is equivalent to *the competitor's support function being dominated by the maintained one's, restricted to the feasible wedge* $\mathcal{F}$. For a rectangle, $h_\phi|_{\mathcal{F}}$ is determined by the two edge values $(c_1,\, c_1 + c_2)$ — which is why two numbers, two probes, sufficed. For a general convex body it is the whole profile $h_\phi|_{\mathcal{F}}$, read off by a continuum of probes. The right object is therefore not two cutoffs but the function $h_\phi|_{\mathcal{F}}$.

5. The one lemma that closes it

Putting Sections 3–4 together: a $\mathcal{C}_{\le\alpha}$ competitor dominates ϕ iff $h_\psi \le h_\phi$ on $\mathcal{F}$, and Lemma 1 says the maintained body is already controlled by its values on $\mathcal{F}$. What remains is to convert "$h_\psi \le h_\phi$ on $\mathcal{F}$, both exact size" into "$A_\psi = A_\phi$." This is a single statement.

Conjecture 2 (reach profile is a complete, rigid invariant)**.** *On exact-size tests in $\mathcal{C}$ with convex acceptance regions, the map $\phi \mapsto h_\phi|_{\mathcal{F}}$ is injective and order-reflecting: if $h_\psi \le h_\phi$ pointwise on $\mathcal{F}$ and both have Gaussian-null-measure exactly γ, then $h_\psi = h_\phi$ on $\mathcal{F}$, hence $A_\psi = A_\phi$.*

The reason I believe Conjecture 2, stated positively:

- **Exact size is a single scalar budget, and lowering reach in one feasible direction must be paid for by raising it in another.** This is exactly what the rectangle case proves in miniature: on $\Phi(c_1)\Phi(c_2) = \gamma$, lowering c_1 forces c_2 up and the diagonal reach $c_1 + c_2$ *strictly* up. So h_ϕ cannot drop on one edge of $\mathcal{F}$ without rising on the other — two exact-size members are never comparable on $\mathcal{F}$. The general statement is the same trade-off run over the whole wedge instead of its two edges.

- **The engine is Gaussian convexity (Ehrhard), which is built for exactly this.** Write $\Gamma(A) := \Phi^{-1}\big(P_0(X \in A)\big)$. Ehrhard's inequality says Γ is concave on convex sets under Minkowski combination,

$$\Gamma\big(\lambda A + (1-\lambda)B\big) \;\ge\; \lambda\,\Gamma(A) + (1-\lambda)\,\Gamma(B).$$

  The constraint "exact size γ" is a level set of Γ, and the feasible reaches $h_\phi(a)$, $a \in \mathcal{F}$, are the supporting-hyperplane positions one moves when deforming A. Ehrhard concavity makes the level set $\{\Gamma = \Phi^{-1}(\gamma)\}$ strictly convex in the reach coordinates, which is precisely the rigidity in Conjecture 2: you cannot weakly enlarge every feasible reach at fixed Gaussian measure. The reverse-hazard/log-concavity computation that powered the rectangle proof is the one-dimensional shadow of Ehrhard.

- **Injectivity of the feasible profile is the convex-geometry analogue of completeness of a sufficient statistic.** $h_\phi|_{\mathcal{F}}$ is, modulo Lemma 1, the full support function of A_ϕ; a convex body is determined by its support function. Order-respecting guarantees the part we cannot probe (outside $\mathcal{F}$) is redundant. So "equal feasible profile" should mean "equal body," and the only freedom left — a uniform inward/outward shift — is killed by the exact-size constraint.

6. The conjectural theorem, and why the application makes it the right one

Assembling: Lemma 1 (proved) reduces everything to the feasible wedge; (1) (a continuum of Lehmann–Romano probes) turns domination into $h_\psi \le h_\phi$ on $\mathcal{F}$; Conjecture 2 (Ehrhard rigidity) turns that into $A_\psi = A_\phi$. That is Conjecture 1.

Remark 1 (naturalness). $\mathcal{C}$ is not abstract. This normal-location problem is the limit experiment for moment inequalities, and the tests that arise there are coordinatewise monotone in the moment functions — that is (M1), the Andrews–Soares monotonicity — while (M2) records which moment binds first, i.e. the priority order on coordinates. A theorem about $\mathcal{C}$ is therefore a theorem about the procedures one actually uses; the short and long tests are its two extreme members, and Conjecture 1 says each is admissible among them.

Remark 2 (why I expect the extensions to go through). Both load-bearing facts are coordinate-free. For $k > 2$ with $\Lambda = \{\mu_k \le \cdots \le \mu_1\}$, (M2) becomes invariance under sorting toward the priority order, Lemma 1 becomes the same statement for the descending rearrangement of a, and $\mathcal{F}$ becomes the wedge between the priority axis and the full diagonal. For $\Sigma \ne I_2$, pass to $\Sigma^{-1/2}X$: the cone, the monotonicities, and the support-sorting all transform affinely, and Ehrhard is affine-invariant. The two-probe picture — coordinate-1 cutoff and diagonal reach — generalizes to "probe each extreme feasible direction," with the number of probes equal to the number of extreme rays of the transformed feasible cone.

One alternative worth keeping in view. If Conjecture 2 proves stubborn to establish in full, the support-function formulation (1) already gives a clean unconditional half: every dominator must satisfy $h_\psi \le h_\phi$ on $\mathcal{F}$ and (by Lemma 1) cannot exceed ϕ outside it either, so a dominator can differ from ϕ only inside the wedge between A_ϕ and its diagonal supporting line. That already localizes any putative dominator to a thin region near the diagonal — the same region where the long test was vulnerable — which is the natural place to run a direct Ehrhard estimate.